1 Example

Below we show a toy example problem that illustrates the usefulness of MEGA. We will use this simple example to explain various aspects of our sampling approach and to contrast our approach with others. This example models an college admission concept that consists of a small number of Boolean predicates. (MEGA also works with fuzzy predicates.)

Suppose Jane plans to apply to a graduate school. Before filling out the forms and paying the application fees, she would like to estimate her chances of being admitted. Since she does not know the admission criteria, she decides to learn the admission concept by induction. She *randomly* calls up a few friends who applied last year and obtains the information shown in Table 1.

Name	*GPA*	*GRE*	*Has Publications?*	*Was Admitted?*
Joe	*high*	*high*	*false*	*true*
Mary	*high*	*low*	*true*	*true*
Emily	*high*	*low*	*true*	*true*
Lulu	*high*	*high*	*true*	*true*
Anna	*low*	*low*	*true*	*false*
Peter	*low*	*high*	*false*	*false*
Mike	*high*	*low*	*false*	*false*
Pica	*low*	*low*	*false*	*false*

Table 1: Admission Samples.

There are three predicates in this problem, as shown in the table. The three predicates are:

- GRE is high,
- GPA is high, and
- Has publications.

The first question arises: "Are all the random samples in Table 1 equally useful for learning the target concept?" Apparently not, for several reasons. First, it seems that $Pica$'s record may not be useful since she was unlikely to be admitted (i.e., her record is unlikely to be labeled positive). Second, both Emily and Mary have the same record, so one of these two records can be redundant. Third, Lulu's record is perfect and hence does not provide additional insight for learning the admission criteria. This example indicates that choosing samples randomly may not produce useful information for learning a target concept.

Now, let us explain how MEGA's sampling method works more effectively than the random scheme. Suppose CCS and QCS are modeled as 2-CNF and 2-DNF, respectively. Their initial expressions can be written as follows:

$$QCS = (GRE = high) \wedge (GPA = high) \wedge (Publications = true) \wedge (GRE = high \vee GPA = high) \wedge (Publications = true \vee GPA = high) \wedge (GRE = high \vee Publications = true).$$

$$CCS = (GRE = high) \vee (GPA = high) \vee (Publications = true) \vee (GRE = high \wedge GPA = high) \vee (Publications = true \wedge GPA = high) \vee (GRE = high \wedge Publications = true).$$

Suppose ψ^* is one. Jane starts by calling his friends whose "profile" fails by exactly one disjunctive term. Jane calls three people and two tell her that they were admitted (i.e., they are the positive-labeled instances) as shown in Table 2.

Based on the feedback, Jane use the positive labeled instances (Joe and Emily) to generalize the QCS concept to $QCS = (GPA = high) \wedge (Publications = true \vee GPA = high) \wedge (GRE = high \vee Publications =$

Round #	*Name*	*GPA*	*GRE*	*Has Publications?*	*Was Admitted?*
1st	*Joe*	*high*	*high*	*false*	*true*
	Emily	*high*	*low*	*true*	*true*
	Dora	*low*	*high*	*true*	*false*
2nd	*Kevin*	*high*	*low*	*false*	*false*

Table 2: MEGA Sampling Rounds.

$true) \ \wedge \ (GPA = high \vee GRE = high)$. At the same time, the CCS is shrunk by using the negative labeled instance (Dora) to $CCS = (GPA = high) \ \vee \ (GRE = high \wedge GPA = high) \ \vee \ (Publications = true \wedge GPA = high)$.

In the second round, Jane attempts to call friends to see if any of the remaining terms can be removed. He calls Kevin, whose profile is listed in the table. Since this sample is labeled negative, the QCS is not changed. But the CCS is reduced to $(GRE = high \wedge GPA = high) \ \vee \ (Publications = true \wedge GPA = high)$.

Simplifying and rewriting both QCS and CCS gives us the following identical expression:

$$QCS = (GPA = high) \ \wedge \ (GRE = high \vee Publications = true).$$

The concept converges and the refinement terminates at this point. We have learned the admission criterion — a high GPA and either a high GRE or publications □